Short Term Trading Model for Asian Equity Index Futures – Using Hurst Exponent

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Abstract

The three-broad behaviour of market prices: trending, mean-reversion and random walk can be studied using the Hurst exponent. The goal of this paper is to develop a comprehensive trading strategy for Asian Equity Index Future by making use of Hurst exponent of the price series. This paper presents research on usages of Moving Hurst Exponent along with other well-known trading indicators to develop a complete short-term trading strategy and trading strategy evaluation experience which the academic and research community can build on.

Keywords: *Equity Index Future Trading, Short Term Trading Models, Moving Hurst Exponent, Mean-Reversion Half-Life, Momentum Regime, Mean-Reversion Regime.*

1. Introduction

Equity Index Futures trading has gained much of attention from trading community since inception and there are good reasons behind the explosive rise of future contracts. These instruments are available on a wide variety of equity indices therefore it can be used with diverse variety of trading strategies. These contracts are available to fit almost any size account. Many of these equity index futures offer options on future contract which facilitates trading strategies to fit different market conditions. These contracts are traded virtually round the clock on electronic platform, hence suitable for quick algorithmic execution. High leverage is another advantage, because a relatively small amount of margin controls a large notional of contract. There is no difference between long contract and short contract form execution point of view, even margin requirements are same. Neither there is any restriction on shorts, nor there is any 'uptick rule' to wait market move before execution of short sell.

These instruments are very liquid, it is easy to place orders quickly while price movements remain fairly stable. The transaction cost for these instruments are significantly cheaper than physical equity. David[1] reflects tax liabilities are less on the gains from equity index futures compared with physical equity. Fontanills & Gentile[2] suggest that unlike some commodity futures, equity index futures are cash settled because it does not involve any physical asset. There are only few other securities that offer the potential for trading that index futures do. These advantages make equity index future appealing for Quantitative trading strategies.

Momentum and Mean-reversion are most common strategies for trading in financial markets. Usually these strategies generate profit when market prices are either trending or mean-reverting because if market prices are random, these strategies would not be effective.

In econometrics, the market price behaviours are classified as black noise (trending), pink noise (mean-reversion) and white noise (random walk). Hurst exponent analysis provides a robust technique to classify market prices into these three categories. Therefore, it can be used to predict and exploit the market price behaviour for short term trading.

Limited academic literature is available on application of Hurst exponent for trading strategies. The other side of the problem is that the research conducted by practitioners and commercial trading firms never comes in the public domain to protect their ability to make money. There is an absence of public research to put together different pieces of empirical evidences on application of Hurst Exponent for a

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complete trading system. This research aims to uncover the under-researched area of studying the effectiveness of Hurst exponent in comprehensive trading strategy development.

The next section of this paper presents reviews literature on Hurst exponent and its application in financial time series analysis. Later part of the paper presents research methodologies and specific techniques.

2. Literature Review

Literature Review for this paper consists of three sections. First section covers literature review of Equity Index Futures trading. Second section covers the interpretation of Hurst Exponent and last section presents its application in financial time series analysis.

2.1 Equity Index Futures Trading

Equity Index as a trading instrument is becoming increasingly popular in recent years. However, Hong & Subrahmanyam[3] suggest that with no physical assets to deal, most equity index trading takes place via index futures and settled in cash. These advantages make it an instrument of choice for quantitative strategies involving for indices or markets of target geography. However, there are not many publications available in academic literature about specific algorithmic or quantitative strategy involving equity index trading. But quite a few publications are available on advantages of trading equity index futures. Kirilenko & Lo[4] argues that due to easy algo automation, Index futures strategies are growing in popularity even among, relatively passive investors even though these instruments are active in nature. Wang et al.[5] presented cross correlation analysis between price return and trading volumes for China Securities Index 300 (CSI300) index futures and concluded that returns and trading volume are long-range cross-correlated. Going by this conclusion, it would be interesting to investigate index futures which are highly liquid and exhibits higher volumes.

2.2 Interpretation of Hurst Exponent

The concept of Hurst exponent is well documented in academic literature. Harold Edwin Hurst, a hydrologist, invented Hurst exponent in 1950's while studying 800 years of flooding records. In 1960's, Benoit Mandelbrot utilized the work of Hurst for financial time-series while modelling long-range dependence of stock returns.

Raimundo & Okamoto[6] argue that over the years, in addition to several other fields Hurst exponent have been applied for the study of verity of financial time series. Similar view has been presented in Kroha & Skoula[7] which says that Hurst exponent provides a measure for long-term memory and fractality of a time series. Since Hurst exponent is robust with few assumptions about underlying system, it has broad applicability for time series analysis.

The value of Hurst exponent (H) ranges between 0 to 1. Qian & Rasheed[8] reflect that based on the value of Hurst exponent, a time-series can be classified into three categories: (1) $H = 0.5$ indicates a random series; (2) $0 < H < 0.5$ indicating an anti-persistent or mean-reverting time series. The strength of mean-reversion increases as H approaches 0; (3) $0.5 < H < 1$ indicates a persistent time series or trend reinforcing, which means the direction of the next value is more likely the same as current value.

Racine[9] argues that Hurst exponent can be interpreted as a measure of trend's strength. As H approaches to 1 the strength of the trend increases. Many academic literatures interpret the mean-reverting or trending behaviour of time-series as long-term memory. Kulish & Vladimír[37] reflect that Hurst exponent of time series more distinct from 0.5 implies large long-term memory and $H = 0.5$ implies absence of long memory means process is random.

A more systematic approach is presented in Kristoufek[10] which argues that Hurst exponent (H) being equal to 0.5 implies either an independent process or a short-term dependent process. Independent

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process has zero auto-covariances at all non-zero lags. On the other hand, short-term dependent process shows non-zero auto-covariances at low lags which decay exponentially to zero at high lags. If $H > 0.5$, auto-covariances of the process are significantly positive at all lags so that the process is called long-range dependent with positive correlations or persistent. Auto-covariances of such process are hyperbolically decaying and non-summable. If $H < 0.5$, auto-covariances are significantly negative at all lags and the process is said to be long-range dependent with negative correlations or anti-persistent. Similarly, to the previous case, auto-covariances of such process are hyperbolically decaying but summable. The persistent process implies that a positive movement is statistically more likely to be followed by another positive movement or vice versa. On the other hand, the anti-persistent process implies that a positive movement is more statistically probable to be followed by a negative movement and vice versa. As Kristoufek[10] provides concrete information on correlation and persistence, this approach makes Hurst exponent an excellent choice for studying complex processes such as the financial time series.

Next section of literature review presents review of literatures on application of Hurst exponents in the study of financial time series.

2.3 Application of Hurst exponents

The application of Hurst Exponent in analysing the behaviour of financial time-series is covered in many interesting publications. Los & Lipka[11] studied daily returns of eight European equity indices and computed Hurst Exponents for each. Study concluded that most indices exhibits long-term dependence on historical time-series and anti-persistent or mean-reverting behaviour. Study further reflected that FTSE is ultra-efficient market with abnormally fast mean-reversion.

Qian & Rasheed[8] present an approach of market predictability by using Hurst Exponent as an input parameter in neural network trained to predict time series. The paper explored the Dow-Jones Index daily return from 1930 to 2004 by taking intervals of 1024 trading days each. Study calculated the Hurst exponent for each 1024 trading days set and found that Hurst exponent for periods with large Hurst Exponents ($H > 0.5$) could be predicted more accurately than periods with random behaviour ($H \approx 0.5$). Their study concluded that only series with large Hurst Exponent can be predicted accurately.

Cajueiro & Tabak[12] calculated median Hurst exponents for return from daily closing prices for 11 emerging market equity indices (Argentina, Brazil, Chile, India, Indonesia, Malaysia, Mexico, Philippines, South Korea, Taiwan & Thailand) as well as indices for the US and Japan. Study concluded that Hurst Exponent are time varying. It further concluded that there is strong evidence of long memory in equity index volatility.

Corazza & Malliaris[13] studied Hurst Exponent of spot returns for five currencies (British Pound, Canadian Dollar, German Mark, Swiss Franc, and Japanese Yen). Study concluded that Hurst Exponent is not fixed but it varies dynamically over time period and Hurst Exponent for sport returns were statistically different from 0.5 in most of the sample studied.

Kristoufek[10] analysed presence of long-range dependence in returns and volatility of Hungarian, Czech and Polish stock indices from 1997 to 2009. Study concluded that there is no significant long-range dependence in returns of these indices. However, the research found a significant long-range dependence in volatility of all three indices. These observations for returns is contradictory with several studies which claim that developing markets are persistent.

Mitra[14] studied daily values of twelve equity index SPI, BSE30, CAC40, DAX, DJI, FTSE100, HSI, KSE, Nikkei, Nasdaq, S&P and STI. Study found that larger Hurst Exponent value if time series is divided into smaller series of 60 trading days. Paper argues that profitable trading strategy can be derived from this behaviour. However, the study did not explore low Hurst Exponent value to check mean reversion pattern.

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Gunay[15] presents fractal structure of the equity market in the Asian countries examining the structure of Japan, Hong Kong, China and Singapore. The paper concluded that there is no evidence of fractality in Japan, but slight sign of fractality in Hang Seng, Shanghai and Singapore. But this paper did not explore the idea of breaking up the data in to different time intervals to see changes in Hurst exponent.

Choudhury & Rajib[16] reflect informational efficiency of India equity market by comparing Hurst Exponent versus Brazil, China, Hong Kong, Japan, Singapore, UK and the USA. This paper did an extensive analysis and concluded efficient market hypothesis across all markets with short-term Hurst exponent closer to 0.5.

Ramos-Requena, Trinidad-Segovia & Sánchez-Granero[17] introduce a new approach to pair trading by using Hurst Exponent as a new method for pair selection. The claim better results than classical distance method and correlation strategies. Results obtained are consistent in reducing the drawdown over the classical ones. The model performs better when the portfolio has ten or more pairs. However, more pair will results in more number of operations therefor it will increase cost.

Kroha & Skoula[7] construct a new Hurst Exponent based technical indicator for trading signals. The performance of new indicator was compared with moving average convergence divergence (MACD) and concluded that new indicator performs better. However, this paper did not go into detailed analysis for complete trading system implementation point of view.

Despite many interesting results found in several studies, the challenge in developing a Hurst exponent based trading strategy is still open.

3. Estimation of Hurst Exponent

A number of methods have been proposed in the literature for calculating the Hurst exponent. Few of the widely used methods are Rescaled Range Analysis (R/S), Detrended Fluctuation Analysis (DFA) and modified GPH model. Out of all different methods, Rescaled Range Analysis (R/S), which was invented by Hurst in 1950's and later popularized by Mandelbrot in 1960's is most popular. Kroha & Skoula[7] suggest that R/S method provides an assessment of how the apparent variability of a series changes with the length of the time-period being considered.

Chan[18] outlines following steps for Hurst Exponent computation using R/S method:

- Divides the time series of length N into 2^k $(k = 0, 1, 2 \dots)$ adjacent subperiods of the same length n, such that $2^k \times n = N$.
- For each sub period of length n, a partial time series is created based on various subperiods, $X = X_1, X_2, \dots X_n$.
- Arithmetic mean $\left(\bar{x} = \frac{1}{n}\sum_{i=1}^{n} X_i\right)$ is calculated and a new mean-adjusted series $(Y_t = x_t - \bar{x})$ column is constructed.
- Cumulated deviations from the arithmetic mean are noted in a new series $(Z_t = \sum_{t=1}^{t} Y_t)$.
- An adjusted range is calculated by $R(n) = \max(Z_1, Z_2, \dots Z_n) - \min(Z_1, Z_2, \dots Z_n)$
- Standard deviation is computed $S(n) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{x})^2}$
- Each adjusted range $R(n)$ is then standardized by corresponding standard deviation $S(n)$, to form the rescaled range $R(n)/S(n)$
- An average rescaled range over all the partial time series of length n is calculated

The process is repeated iteratively using $k = 0, 1, 2 \dots$ for each length $n = N/2^k$ of subperiods.

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Peters[19] presents a more formal formula for the estimation of the Hurst exponent as:

$$E\left[\frac{R(n)}{S(n)}\right] = Cn^H \; as \; n \to \infty$$

$R(n)$ is adjusted range

$S(n)$ is its standard deviation

$E[x]$ is the expected value, which is arithmetic average

n corresponds to time span of the observations

C is a constant and H is the Hurst exponent

The Hurst exponent H is the slope of the plot of each range

$\log [R(n)/S(n)]$ as a function of $\log n$.

This paper makes use of R/S method for Hurst Exponent computation. Next section of report presents research methodology adopted for the paper.

4. Research Methodology

The proposed research is based on back-testing methodology (described in next section) to test the trading strategy and study its effectiveness. Tortoriello[20] argues historical performance of a trading strategy provides a statistical estimate of future performance. Emphasis of this research is to study the viability of strategy by testing it with historical data. The performance of trading strategy is analysed by performing back-testing. Following section presents various aspects of back-testing.

4.1 Back-testing of Trading Strategy

Haas[21] points out that back-testing gives significant insight about trading strategy and dynamics of the trade. Tortoriello[20] defines back-testing as a statistical analysis on historical data to determine whether employing a given trading strategy will results in excess returns over time. Chan[22] argues that a key difference between a traditional investment management and a quantitative investment is that quantitative investment strategy can be subjected to back-testing to evaluate the past performance. Chan[22] further says that even if a trading strategy description has complete detail with all the historical performance data, it's still advisable to perform back-testing. Back-testing of the research ensures clear understanding of strategy and all the details of implementation. It also allows for experimentation with variations of the original strategy, thereby refining and improving the strategy. Back-testing provides with filtration techniques – thus, strategies that do not meet specific requirements can be eliminated. Certain market phenomena such as transaction costs, order routing latency, liquidity, or any other micro market structure can be tested under a safe environment of back-testing. Strategy optimization, though fraught with biases, could also be well catered during back testing. Halls-Moore[23] suggests back-testing also ensures that a strategy has been correctly implemented and the performance could be replicated.

While back-testing does not allow one to predict how a strategy will perform in future, but it provides a clear understanding of vulnerabilities of a strategy through a simulated encounter with real-world conditions of the past. Hence the designer of the trading strategy can 'learn from the history' without actually spending money.

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4.1.1 Event Driven Back-testing Framework

As Halls-Moore[23] outlines event driven trading systems are widely used in live algorithmic trading. It can be made to simulate historical data to mimic live trading environment. It allows back-testing of strategies in a manner that is extremely similar to live execution. It can run for a specific period after an initial setup and executes its trading logic when specific event occurs. These events can be driven by live data feed or historical data. It can connect back-testing to historical market data and simulated broker whereas live trading strategy to real-time market data feed and broker. Historical market data scanned by system triggers events to generate trading signal and simulated execution of trade whereas live Market data feed received by system triggers events to generate new trading signal and real execution of trade. Such a system executes in a continuous loop waiting to receive events and handle them appropriately. There are many open source event driven trading systems are available like ZipLine, PyAlgoTrade, BackTrader etc. Due to its simplicity and easy installation, PyAlgoTrade framework has been picked for this paper.

4.2 Performance Measurement

Performance measure allows review of effectiveness of a trading strategy, academic literature as well as commercial reports mention quite a few performance measures for trading strategies. Halls-Moore[23] outlines without assessment of performance, it is difficult to determine if the trading strategy performance is due to luck or due to actual edge. Chan[18] & [22] outlines that variety of performance measures are used to assess a trading strategy. Average Annual Return, Compounded Annual Growth Rate, Sharpe Ratio and Maximum Drawdown, Maximum Drawdown Duration and Calmar Ratio are key performance metrics for performance of a trading strategy.

4.2.1 Cumulative Return

Chan[22] argues that average annual return is commonly used to compare the returns generated from trading strategy under evaluation. To facilitate comparison across different trading strategies often return is annualized from the total return data obtained from back-testing over a period of time. Chan[22] argues that average annual return ignores compounding. Compounded Annual Growth Rate (CAGR) is not the same as average annualized returns. Given money is not withdrawn after each trading day, simple return makes less sense. Compounded Annual Growth Rate (CAGR) is rate of return a portfolio would earn from the beginning to ending balance, assuming profits were reinvested during the lifespan of the investment. Murphy[24] points out that CAGR ignores volatility and assumes growth during the time frame was steady.

$$\text{CAGR} = \left(\frac{\text{Ending Value}}{\text{Starting Value}}\right)^{\frac{1}{\text{n}}} - 1$$

$$n - \ is\ number\ of\ years$$

Another popular measure Cumulative Return is measure of the aggregate amount that trading strategy has gained or lost over time, independent of the period of time involved.

$$Cumulative\ Return = \frac{Current\ Price - Strating\ Price}{Starting\ Price}$$

Cumulative returns are expressed in percentage and due to its simplicity, its widely used in Back-testing. This paper makes use of cumulative returns for the performance measurement.

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4.2.2 Sharpe Ratio

Sharpe Ratio is a widely used risk-adjusted return measure and it takes risk-free return as its benchmark Chan[18]. Since, there is no risk on a risk-free asset, standard deviation of risk free return is zero.

$$Sharpe\ Ratio = \frac{Average\ of\ Excess\ Returns}{Standard\ Deviation\ of\ Portfolio\ Returns}$$

$$Excess\ Returns = Portfolio\ Returns\ - Risk\ Free\ rate\ of\ Return$$

Chan[18] argues that Sharpe ratio facilitates performance comparison across different strategies. Sharpe ratio is usually low if a strategy trades only few times a year and has a deep or lengthy drawdown. Sharpe ratio has been considered as performance measure for this paper.

4.2.3 Maximum Drawdown

Maximum drawdown focuses on capital preservation[25]. It approximates worst-case scenario for a trading strategy for a period. It is the maximum loss observed from peak to trough, for a portfolio, before a new peak is attained. It helps to measure the risk incurred by trading strategy and helps to determine if its practical to take on such risk. Miner[26] argues capital preservation is key to long-term successful trading and an important criterion for most investors. Chan[18] states that it is a better measure of tail risks, as it does not rely on measuring just the second moment of returns distribution. It is also asymmetric i.e. not concerned about periods with extremely positive returns.

$$Maximum\ Drawdown = \frac{Trough\ Value - Peak\ Value}{Peak\ Value}$$

From trading strategy development point of view, it can be visualized during back-testing phase. As this research involves back-testing, Maximum Drawdown has been considered as one of the measures of performance for this research.

4.2.4 Maximum Drawdown Duration

Maximum Drawdown Duration is an extension of Maximum Drawdown. It explains the length of time (in days, weeks, months or years) the portfolio was in max drawdown. Maximum Drawdown Duration indicates the longest period of time it takes for the portfolio to recover from losses. A short maximum drawdown duration means that portfolio quickly recovers from losses. Kazemi et al[27], points out that a large maximum drawdown duration could also reflect that effective risk controls are not in place or the trader is slow to react to changing market conditions.

4.3 Validation of Trading Strategy

Fabozzi, Focardi & Kolm[28] argues that though back-testing methodology provides significant insight about trading strategy but it has many drawbacks. Back-testing based trading strategy development is susceptible to many overfitting drawbacks like data mining bias, look-ahead bias, data-snooping bias etc. Chan[22] argues that an overfitted trading strategy work fabulously with data-set used in development and fail to perform with other data-set. Chan[22] further says that look-ahead bias result from use of data during historical back-testing that were not actually available at the time of trading. Data-snooping bias arises from over optimization of parameters of the back-test based on transient noise in the historical data. It often results into inflated relative performance strategy undergoing the back-

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testing. Given the veracity and impact of these biases, Validation of trading strategy is critically important. In academic literature, Out of Sample Tests and Robustness Tests are widely used methods to validate trading strategies:

4.3.1 Out of Sample Tests

Out of Sample test is the most popular validation method and it requires setting aside a portion of the data available before using it in development and testing the trading strategy. After completion of development and back testing of the developed trading strategy, the portion of data set aside can be used to obtain an unbiased estimate of performance. This paper makes use of out of sample test to validate the developed trading strategy.

4.3.2 Robustness Tests

Robustness test involves observing the performance of trading strategy while forcing small variation in parameters, entry and exit conditions for long or short. The goal of this test is to determine that small variation in strategy parameter and trading condition has low impact on performance measures like cumulative return or maximum drawdown as suggested by Halls-Moore[23].

5. Historical Data

As discussed in last section, this paper involves back-testing of trading strategy using historical market data. Historical daily price data series for Equity Index Futures has been sourced from Investing.com and formatted in the appropriate format. Taylor[29] suggests that at least four years of daily price data (or more than 1,000 observations) are often required to obtain interesting results. However, if feasible, time series of eight or more years (more than 2,000 observations) should be analysed.

Last 10 years daily price data has been considered for this research for following Asian Equity Index Futures: Japan's Nikkei, South Korea's KOSPI, Australia's ASX, India's Nifty, Indonesia's JKSE, Hong Kong's HSI & HSCEI and Taiwan's TSEC. Only highly liquid future contracts have been considered. For Japan, Korea and Australia prices are from quarterly future contract whereas for rest all markets prices are from monthly future contract (Appendix I summarizes the mechanism for various Asian equity index futures contract).

Schwager[30] reflects that the transitionary life of futures contracts is one of the major obstacles for back testing trade ideas. Chan[31] states that futures contract has expiry dates therefore a trading strategy involving futures is a trading strategy on many different contracts.

Most historical data vendors for futures, including the one used in this paper, recognize the issue of transitionary life across futures contract and thus provide a continuous contract data.
Usually the Nth Nearest contract, which is an artificial continuous data series representing a concatenation of price for successive contracts over time, of the Nth nearest future. The gaps or jumps between contracts are left unchanged in this data series. Masteika[32] argues that short term trading such series is satisfactory.

6. Trading Strategy Development Approach

This research leverages the idea of exploring well-known trading indicators and strategies to complementary the development of a new Hurst Exponent based trading model. As Halls-Moore[23] argues with any single quantitative or technical indicator it can be challenging to form a profitable inter-day trading system. Echoing these views multiple indicators has been considered in this paper. It aims to apply quantitative and technical methods to explain the observed behaviours in the historical data

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while attempting to create a realistic inter-day trading strategy. Following are the key indicators studied during the development of comprehensive trading system:

1. Moving Average Indicators
2. Momentum Indicators
3. Mean Reversion Indicators
4. Moving Hurst Exponent

In addition to exploring various trading indicators, research does a Moving Hurst Exponent analysis for all the index under study. The analysis is performed with different look back period to identify any interesting pattern which can be used in the development of trading strategy.

6.1 Moving Average Indicators

Pring[34] outlines that Moving Average (MA) attempts to tone down fluctuations in price by generating a smoothed curve with minimum distortions. It measures the direction of existing trends and can be used to determine the trend changes, support and resistance levels. It can be calculated for any time period. On arrival of new data point in the time series, new data point is added in average computation and the first data point is removed. This process is repeated and average is taken at each step. Since the average moves with each passing time period, it is called a 'moving' average. Usually 50-day MA line (referred as Shorter/Fast line) and 200-day MA line (referred as Longer/Slow line) together is used for analysis. Murphy[33] reflects, usually buy or sell signals are generated based on cross-over of these fast and slow line.

6.1.1 Simple Moving Average

Simple Moving Average (SMA) is created by taking average of all the observations in the look-back period. Murphy[33] argues that this indicator is called 'simple' because each observation is given equal weight in the calculation. SMA is one of the most widely used classic technical indicator. Usually two SMA with different look back period are taken together to analyse market trends.

6.1.2 Exponential Moving Average

In contrast to SMA, Exponential Moving Average (EMA) reduces the lag by applying weight to recent observation. The weighting applied to the most recent observation depends on the number of periods. A given days' EMA calculation depends on the EMA calculations for all the days prior to that day. For a reasonably accurate n-days EMA computation, its required to have far more than n-days of data. Pring[34] advises on three steps to calculating EMA:

- First, calculate the simple moving average for the initial EMA value. As EMA has to start somewhere, usually a simple moving average is used as the previous period's EMA in the first calculation.
- Second, calculate the weighting multiplier.
- Third, calculate the exponential moving average for each day between the initial EMA value and today, using the price, the multiplier, and the previous period's EMA value.
- And process is continued until the end of series.

$$Initial\ SMA = \frac{n - period\ sum}{n}$$

$$Multiplier = \frac{2}{n+1}$$

$$EMA = \left(Close\ Price\ -\ EMA_{Previous\ day}\right) * multiplier\ +\ EMA_{Previous\ day}$$

EMA is widely used for the purpose of identifying reversal in trends.

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6.2 Momentum Indicators

When prices are trending up or down, it is referred to have momentum. The moving averages are extremely useful indicators, but they identify a change in trend only after it has taken place. This is where significance of momentum indicators come into play. Momentum indicators can detect the latent strengths or weaknesses in price time series often well ahead of the final turning point. This section presents some of the widely used momentum indicators which has been experimented as part of this study.

A price chart reflect prices are either rising or falling. Murphy[33] reflects an oscillator chart states more about the momentum of a market. An oscillator shows the rate of change at which a market is rising or falling. Such oscillator hints whether the current trend is gaining or losing its momentum.

6.2.1 Rate of Change

As its name would suggest Rate of Change (ROC) is rate of price changes over a given period of time.

$$n\ week\ ROC\ at\ time\ t = \frac{Current\ Price\ at\ time\ t}{Price\ n\ week\ ago\ from\ time\ t}$$

ROC time series oscillates around a central reference point. Pring[34] states rising ROC implies expanding momentum whereas falling ROC implies a declining momentum. Usually multiple period ROCs like 1-week line, 2- weeks line, 5-weeks line, etc. is used together for analysis. The midpoint line is the key to ROC oscillator. A crossing above the midline is considered to be positive and below the midline is negative. Murphy[33] reflects ROC can also be used to spot the extremes in a market. When the ROC lines are too far above or too far below the mid-line, the market is considered to be overbought or oversold respectively. If the ROC line starts to move back toward the midline, it can be an early signal that the current trend is losing momentum.

6.2.2 Relative Strength Indicator

Relative Strength Index (RSI) is measure of speed of price movements. RSI oscillates between zero and 100. Readings over 70 are considered to be overbought and below 30 reflect to be oversold. Applying these two boundaries to price time series simplifies the extreme indicators. Murphy[33] argues that the midpoint value of 50 can serve the same purpose as the zero line and crossings above and below that value can generate trend signals. Pring[34] states RSI also provides a constant trading band for comparison purposes. RSI above its overbought zone or below its oversold zone can indicate a potential upcoming reversal. The range for RSI varies depending on RSI parameters, strength of trend and volatility of the underlying security. Brown[35] advises that a bull market range for RSI tends to fluctuate between 40 and 90, with the 40-50 zones acting as support where as in a bear market RSI tends to fluctuate between 10 and 60 with the 50-60 zone acting as resistance.

RSI is calculated in following steps:

Step 1: Calculate Average Gain and Average Loss
First calculations for average gain and average loss are simple x-period averages:

$$First\ Average\ Gain\ =\ Sum\ of\ Gains\ over\ the\ past\ x\ periods\ /\ x$$
$$First\ Average\ Loss\ =\ Sum\ of\ Losses\ over\ the\ past\ x\ periods\ /\ x$$

The second, and subsequent, calculations are based on the prior averages and the current gain/ loss

$$Average\ Gain\ =\ [(previous\ Average\ Gain) \times (x-1)\ +\ current\ Gain]\ /\ x$$

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$$Average\ Loss\ =\ [(previous\ Average\ Loss)\ \times (x-1)\ +\ current\ Gain]\ /\ x$$

Step 2: Calculate RS

$$RS\ = \frac{Average\ gains\ over\ past\ x\ periods}{Average\ losses\ over\ past\ x\ periods}$$

Step 3: Calculate RSI

$$RSI = 100 - \frac{[100]}{1+RS}$$

Taking the prior value plus the current value is a smoothing technique similar to that used in calculating an exponential moving average. Usually 14 days period is used for RSI calculation however it can be lowered to increase sensitivity or raised to decrease sensitivity.

6.2.3 Stochastic Oscillator

Murphy[33] states Stochastic indicator also referred as Stochastic Oscillator is a measure of level of current close price relative to highest high price and lowest low price over a given period of time. He further advises that the value of Stochastic Oscillator is above 50 when the close price is in the upper half of the range and below 50 when the current close price is in the lower half. Value below 20 indicate that current price is near its low for the given time period. High readings (above 80) indicate that current price is near its high for the given time period. As it follows the speed or momentum of a price, which can be used to forecast reversals. Murphy [33] points out that it can be used to identify overbought and oversold levels as its value oscillates between 0 and 100. Usually value 80 is considered overbought threshold and 20 as the oversold threshold. It has got two lines, slower %D line is a moving average of the faster %K line. Trading signals are generated when the two lines cross or when their value is > 80 or < 20.

$$Fast\ line\ (\%K) =\ (\ Close\ -\ Lowest\ Low)/\ (Highest\ High\ -\ Lowest\ Low)\ \times\ 100$$

$$Slow\ line\ (\%D) =\ 3\ day\ SMA\ of\ \%K$$

$$Lowest\ Low\ =\ lowest\ low\ for\ the\ look-back\ period$$

$$Highest\ High\ =\ highest\ high\ for\ the\ look-back\ period$$

$$\%K\ is\ multiplied\ by\ 100\ to\ move\ the\ decimal\ point\ two\ places$$

Murphy[33] says that divergences occurs when a new high or low in price is not confirmed by the corresponding high or low in the value of Stochastic Oscillator. A bullish divergence forms when price records a lower low, but the Stochastic Oscillator forms a higher low. This shows less downside momentum and could predict a bullish reversal. A bearish divergence forms, when price records a higher high, but the stochastic oscillator forms a lower high. This shows less upside momentum and could predict a bearish reversal. A bearish divergence can be confirmed when Stochastic Oscillator break below 50 (the centreline), similarly a bullish divergence can be confirmed when Stochastic Oscillator break above 50.

6.2.4 Moving Average Convergence Divergence

Murphy[33] further points out that usually MA outperforms other indicators during trending period and oscillators outperforms other indicators during choppy market periods. Moving Average Convergence Divergence (MACD) combines the best of both MA and oscillators. It generates trend-following signals as well as helps to determine if trend is overbought or oversold. It involves three MA in its construction. The first line (called the MACD line) is the difference between two exponentially

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smoothed moving averages of the price (usually 12 and 26 periods). The longer average (26 periods) is subtracted from the shorter (12 periods) to obtain the MACD line. A moving average (usually 9 periods) is used as signal line in the analysis.

$$MACD = Shorter\ Period\ EMA - Longer\ Period\ EMA$$

$$Signal\ line = Signal\ period\ EMA$$

$$MACD\ Histogram = MACD\ line - Signal\ line$$

Miner[26] says as its name would suggest MACD represents convergence and divergence of the two EMA. Convergence occurs when the two EMA move towards each other. Divergence occurs when the two EMAs move away from each other. Miner further points out that the shorter EMA is faster and responsible for most MACD movements. The longer EMA is slower and less reactive to price changes. Positive MACD indicates that the shorter EMA is above the longer EMA. Positive values increase as the shorter EMA diverges further from the longer EMA. This means upside momentum is increasing. Negative MACD values indicate that the shorter EMA is below the longer EMA. Negative values increase as the shorter EMA diverges further below the longer EMA. This means downside momentum is increasing.

Miner[26] reflects, when MACD turns up from below centreline to become positive, it is considered as bullish. When it turns below the centreline to become negative, it is considered bearish. Centreline crossovers can last a few days or a few months, depending on the strength of the trend. On the other hand, in case of sustained downtrend, MACD remains negative.

Pring[34] states that, when the MACD line crosses from below to above the signal line, it is considered as bullish, however when the MACD line crosses from above to below the signal line, the indicator is considered bearish. The MACD histogram is positive when the MACD line is above signal line and negative when the MACD line is below the signal line.

6.3 Mean Reversion Indicators

Chan[31] argues that mathematical description of a mean-reverting price series is that the change of the price series in the next period is proportional to the difference between the mean price and the current price. It further says that if a price series is mean reverting, then the current price level will signal the future price move. If price level is higher than mean, the next move will be a downward move, if the price level is lower than the mean the next move will be an upward move.

6.3.1 Augmented Dickey-Fuller Test

Chan[31] points out that Augmented Dickey-Fuller Test (ADF) is based on the observation that if price is mean reverting then, a price level higher than mean should have downward move next; and if price is lower than mean, then its next move should be upward move.

Chan[31] further describes price changes using a linear model:

$$\Delta y(t) = \lambda y(t-1) + \mu + \beta t + \alpha_1 \Delta y(t-1) + \cdots + \alpha_k \Delta y(t-k) + \epsilon_t$$

Where, $\Delta y(t) \equiv y(t) - y(t-1)$, $\Delta y(t-1) \equiv y(t-1) - y(t-2)$, and so on.

The ADF test will check if $\lambda = 0$. If the hypothesis $\lambda = 0$ can be rejected, it means the next move $\Delta y(t)$ depends on the current level $y(t-1)$, and therefore it is not a random walk. The test statistic is the regression coefficient λ, divided by the standard error of the regression fit:

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$$\frac{\lambda}{SE(\lambda)}.$$

Chan[31] argues $\frac{\lambda}{SE(\lambda)}$ has to be more negative than the critical value for the hypothesis to be rejected. The critical values depend on the sample size and whether the price series has a non-zero mean $-\frac{\mu}{\lambda}$ or a steady drift $-\frac{\beta t}{\lambda}$. In practical trading, the constant drift in price, tends to be of a much smaller magnitude than the daily fluctuations in price. So, for simplicity the assumption is that drift term is zero ($\beta = 0$).

6.3.2 Half-Life of Mean Reversion

Chan[31] points out that Half-life can be used to determine optimal holding period for a mean-reverting process. Half-Life estimates how long does it take for a time series to revert back to half its initial deviation from mean. This can also be used as a selection criterion for screening whether the trade is suitable for short-term trading.

The mean reversion of a time series can be modelled by the Ornstein-Uhlenbeck stochastic differential equation as:

$$dX_t = \theta\,(\mu - X_t)dt + \sigma dW_t$$

Where, X_t is the mean-reverting process, μ is the mean value over time θ is speed mean reversion and dW_t is Gaussian noise. With time series of X_t linear regression fit can be performed to find the θ. Using Ito Lemma, the equation can be solved explicitly to deduce that average value of X_t follows an exponential decay to its mean μ, and the half-life of this exponential decay is:

$$\frac{\ln(2)}{\theta}$$

which is the expected time it takes for the spread to revert to half its initial deviation from the mean. Thus, as Chan[31] reflects, larger the θ (speed of mean reversion), shorter the half-life and thus stronger the mean-reversion.

6.3.3 Bollinger Bands

Bollinger Bands are volatility/standard deviation bands placed above and below a moving average line. The band widen when volatility increases and contract when volatility decreases.
Time span and level of standard deviation are two parameters of Bollinger Bands. Time span 20 days and 2 standard deviation are recommended parameters.

$$Middle\ Band = 20 - day\ simple\ moving\ average\ (SMA)$$

$$Upper\ Band = 20 - day\ SMA + (20 - day\ standard\ deviation\ of\ price * 2)$$

$$Lower\ Band = 20 - day\ SMA - (20 - day\ standard\ deviation\ of\ price * 2)$$

7. Trading Model Parameter Optimization

As discussed in previous sections, various trading indicators involve parameters like look-back period. These parameters should be optimized in a systematic fashion to improve the performance. Halls-Moore[23] suggest that comprehensive trading model will often have many parameters and different measures for performance. Therefore, a decision on which set of parameters to optimize and when to optimize would be required.

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Another aspect of parameter optimization is overfitting. Chan[22] argues overfitting of a model or trading strategy is one of the bigger dangers in parameter optimization. This occurs when performance degrades substantially on 'out of sample' data, but performs well on trained 'in sample' data. Overfitted trading strategy can perform extremely well during back-testing but when deployed on live trading, the strategy becomes unprofitable. In other words, a strategy too closely fit to a limited set of data points might not work properly outside of those data points. Keeping these into prospective, this paper makes use of 'out of sample' validation to ensure right amount of optimization of parameters.

Halls-Moore[23] reflects like any other discrete mathematics optimization problem, 'multiple parameter optimization' can increase computational complexity by orders of magnitudes, which can be computationally very expensive. This paper makes use of observed back-testing results to optimize the trading model parameters. The optimization methods by PyAlgoTrade platform has been used in this paper to decide the optimal parameters.

8. Trading System Implementation

This section presents the full implementation of trading strategies using the event-driven back-testing framework PyAlgoTrade. This paper follows Jansen[36] which argues any successful algorithmic trading strategy is driven by combination of indicators. The approach taken in this paper involves multiple trading indicators. It follows the classic investment portfolio approach where diversification is achieved by investing in different assets. The assumption is that using combination of strategies together will result in strategy diversification which will make trading system more resilient. The trading indicators described in the previous sections of this report forms basis of the implementation. As reflected by Chan[22] the two basic categories of trading strategies are mean-reverting and momentum strategies. Momentum is also referred as trending. But as outlined in many academic literatures, periods of mean-reversion and period of momentum behaviour are two different regimes of market. Therefore, for a successful trading strategy its critical to determine the regime of market out of mean-reversion regime and momentum regime.

As discussed in previous sections, the Hurst exponent can be used to determine whether the price series is mean-revering or exhibiting momentum behaviour. Building on this knowledge, paper introduces Moving Hurst Exponent indicator to identify and exploit the regime switching between mean-reversion regime and momentum regime. Following a successful determination of regime, a trading system can be programmed to follow strategy suitable for specific regime. Therefore, the three-key task for this strategy implementation evolves to identification of regime, trading strategy for mean-reversion regime and trading strategy for momentum regime. Next section of this report presents Moving Hurst Exponent indicator.

8.1 Moving Hurst Exponent Indicator

As discussed in previous section, value of Hurst Exponent can be used to determine the regime of market in between Mean Reversion regime and Momentum regime. It's essential to have a robust 'Moving Hurst Exponent' indicator to be able to experiment with different look back periods for Hurst Exponent computation. After several trial-and-error, a new indicator 'Moving Hurst Exponent' was implemented in PyAlgoTrade framework (source code detail Appendix II). It can be used to determine the value applicable Hurst Exponent at a specific point of time.

As discussed in previous section, the most popular method for Hurst exponent computation is Rescaled Range Analysis (R/S) method. Hence it has been picked for Hurst Exponent Computation. The Moving Hurst Exponent indicators take period as parameter and work on the price time series to determine the value of Hurst Exponent. If period parameter is n-days it implies that data points for Hurst Exponent calculation comprises last n-days data from time-series. As PyAlgoTrade is an event driven framework, each arrival of new data-point constitutes an event. Every arrival of new event adds the latest new data point in the executing code of Moving Hurst Exponent indicator and removes the oldest data point. With change in set of data points, Hurst Exponent is re-calculated on every arrival of new

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event. As all other parameter changes due to changing market conditions, its logical to allow trading system to adopt to evolving Hurst Exponent of time series to capture profit more quickly and stop losses quickly. Next section of this report presents the application of 'Moving Hurst Exponent' in trading strategy development.

8.2 Determination of Regime

The concept of market regime is most basic to financial markets. With the desire to model regime switching between mean-reverting and momentum regimes for a successful trading system implementation, several experiments have been done in this paper. Using the new indicator 'Moving Hurst Exponent', the regime of price time series out of mean reverting regime and momentum regime can be determined for certain time horizon. Like any inter-day trading strategy, a relatively shorter lookback period will always result with the indicator making reversals very close to the price-swing highs and lows. Therefore, only longer than week period has been considered for mean reverting or momentum regime specific strategy development and Multiple weeks period has been considered for Moving Hurst exponent computation. Next section of this report presents another indicator which has been implemented in PyAlgoTrade framework to exploit mean-reverting regime effectively.

8.3 Mean Reversion Half-Life Indicator

Going by Chan's[31] argument that frequency and duration of mean reversion is critical for profitability of a trading strategy, half-life of mean reversion has been experimented in this paper. The idea is to determine and use half-life of mean reversion to generate buy signal near lowest price point of mean reversion cycle but sell signal near highest price point of mean reversion cycle.

The implementation follows the method discussed in section 6.3.2, the speed of mean reversion θ is computed by performing the linear regression fit of time series and half-life is computed using formula:

$$\frac{\ln(2)}{\theta}$$

It has been implemented using PyAlgoTrade framework (source code detail Appendix II)

Theoretical assumption is that if half-life period is relatively smaller, strategy would be able to complete many round-trips and hence many profitable trades. Buy or Sell signals are often determined using multiple moving averages lines however this indicator makes use of one moving average and a moving standard deviation of price. Along with half-life period, moving average and moving standard deviation of price has been taken into consideration to determine peak and trough price level. Usually look-back period for moving average or moving standard deviation are parametrised but this paper follows Chan[31] which argues that setting the look-back period equal to the half-life is close to optimal.

$$look\ back\ period\ for\ moving\ standard\ deviation\ =\ half\ life\ period$$

$$look\ back\ period\ for\ moving\ average = half\ life\ period$$

Using above mentioned equation, 'Mean Reversion Half-Life' indicator has been implemented in PyAlgoTrade framework (source code detail Appendix II). Next section of report presents the strategy implementation for Mean-Reversion regime.

8.4 Strategy for Mean-Reversion Regime

Once trading system has discovered the regime as mean-reverting, the next task is to find a strategy that works well with price series exhibiting mean-reversion. Starting with newly developed 'Mean

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Reversion Half-Life' indicator, many different methods were evaluated in this paper to formulate an effective strategy for mean-reversion. This report presents two strategies named as 'Attempted Approach' and 'Concluded Plan'. Next section presents the attempted approach.

8.4.1 Attempted Approach

Initially half-life of mean reversion-based strategy was attempted for mean-reversion regime. The idea was to determine half-life to decide the peak and trough price level of period under examination and use half-life period to automate buy or sell decision. The theoretical assumption was that using moving half-life information it would be possible to buy at lower price point and sell at higher price point of mean reversion cycle.

Implementation Steps:

- Once trading system has determined that price series is mean reverting using 'Moving Hurst Exponent' indicator, the next task is to determine half-life of mean reversion.
- The newly developed 'Mean Reversion Half-Life' indicator provides half-life of mean reversion and it also provides moving average and moving standard deviation which are calculated by taking half-life period as look-back period.
- Trading signal is generated based on price divergence by multiples of moving standard deviation from moving average.
- As moving average and moving standard deviation change due to changing market conditions, its logical to allow trading system to adopt to evolving deviation of price to capture profit more quickly.

$Condition\ for\ Buy\ order$:

$$price < moving\ average\ of\ price\ - n * moving\ standard\ deviation\ of\ price$$

$Condition\ for\ Sell\ order$:

$$price > moving\ average\ of\ price\ - n * moving\ standard\ deviation\ of\ price$$

$Where\ n\ is\ calibrated\ parameter\ in\ the\ range\ 0.5\ to\ 1.5$

- During experimentation with historical price data listed in previous section, it was found that 'n' value from 0.5 to 1.5 gives optimal results.

The reasoning behind usage of moving average and moving standard deviation in the above formulation is that it changes due to changing market conditions therefore its logical to allow trading system to adopt to evolving deviation of price in order to capture profit rapidly.

Using above mentioned logic trading strategy has been implemented in PyAlgoTrade framework (source code detail Appendix II).

Due to complication in calibration of parameter and unstable performance across different indices, this approach was not pursued after initial back-testing. Next section of this report presents concluded trading strategy for the mean reversion regime.

8.4.2 Concluded Plan

After several attempts, it was realized that a simple Relative Strength Index (RSI) along with fast SMA and slow SMA gives a better performance for treading strategy in the mean-reversion regime. As described in previous section, RSI is a measure of speed of price movements and it oscillates between zero and 100. As Murphy (2009) argues value 50 for RSI can serve the midpoint of a range bound price time series, it can be considered as the approximate mid-point of mean reversion cycle. The evolved

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trading strategy also follows Miner[26], which tried to predict the reversals from bullish to bearish or from bearish to bullish by making use of two SMAs with different look-back periods. Slow SMA which has longer look-back period acts as entry indicator, whereas fast SMA which has shorter look-back period acts as exit indicator. It's worth to mention, Miner[26] further argues that while using dual time frame indicator to predict the reversals, lager time frame/slow-line can be used to identify trade direction and a smaller time frame/fast-line for trade execution. The evolved trading strategy, makes use of it to form key trading signals when the indicator reverses from bullish to bearish or from bearish to bullish as follows:

Trade Entry:

- If the price crosses above slow (or entry) SMA, results in bullish momentum reversal, therefore a buy signal.
- If the price crosses below slow (or entry) SMA, results in bearish momentum reversal, therefore a sell signal.

Trade Exit:

- If the price crosses below fast (or exit) SMA, close out buy positions
- If the price crosses above fast (or exit) SMA, close out sell positions

As Chan[22] argues, due to increasing competition from traders, the mean-reverting effect typically gradually eliminates the opportunity for large profit. This is where RSI can come into play. In order to exploit this diminishing return, evolved trading strategy identifies the floor and upper bound just in time by making use of RSI value. The value of RSI has been used to for early identification of stage of mean reversion.

Following logic has been coded in the strategy:

- Enter Long signal confirmation if RSI is less than oversold threshold
- Enter Short signal confirmation if RSI is greater than overbought threshold

The idea behind supplementing the dual time frame indicator with RSI oversold and RSI overbought threshold is to ensure that Long and short signal are on the opposite side of mid-point of mean reversion cycle. Using above mentioned logic trading strategy has been implemented in PyAlgoTrade framework (source code detail Appendix II). Next section of report presents the strategy implementation for Momentum regime.

8.5 Strategy for Momentum Regime

As Chan[31] outlines price series momentum means that past returns are positively correlated with future returns. Therefore, an algorithmic strategy would require calculation of correlation coefficient of the returns. For computing the correlation coefficient, we will have to pick a specific time lag for the returns. Chan[31] further says that often most positive correlations are between returns of different lags. Therefore, its critical to find the optimal pair of past and future periods that gives the highest positive correlation and use that as look-back and holding period for momentum strategy.

By presenting empirical evidence Miner[26] argues that momentum indicators do not represent price trends. Price and momentum do not always trend together. For example, a momentum indicator may make a bearish reversal and decline while price trend continues to advance. It may be because rate of change of the price trend is decreasing even through price continues to advance. The bullish trend is just slowing down resulting into bearish momentum indicator even though price trend continues to bullish.

8.5.1 Attempted Approach

Initially classic Bollinger Bands technique along with a slow and a fast SMA line was tried for the momentum regime. Typical Bollinger Bands based strategy generates a buy signal when price is below

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lower bank and a sell signal when price breaks above the upper band. It just replies on identifying prices that are deviated from the mean. However, in the attempted strategy in addition to Bollinger Bands, dual SMA line has been used to identify the momentum reversal as per Miner[26].

Therefore, following logic has been coded in the strategy:

- If the fast SMA crosses above the slow SMA, indicates the bullish reversal of momentum/trend therefore a buy signal
- If the fast SMA crosses below the slow SMA, indicates the bearish reversal of momentum/trend therefore a sell signal

These momentum cross over has been taken in addition to Bollinger Bands.

- Buy signal only if price is below Bollinger Band's lower band
- Sell signal only if price is above Bollinger Band's upper band

In this paper the time span for Bollinger Bands has been taken as parameter in the range of 15 to 25 as the recommended time span for Bollinger Bands is 20. The standard deviation for Bollinger Bands is parameterised and trial has been done with 2 standard deviation band.

Using above mentioned logic trading strategy has been implemented using PyAlgoTrade framework (source code detail Appendix II) in this paper. But due to unstable performance across different indices, this approach was not pursued after initial back-testing. Next section of this report presents concluded trading strategy for the momentum regime.

8.5.2 Concluded Plan

Moving Average Convergence Divergence (MACD)'s signal line cross-over has been considered as final strategy for momentum regime. Cross over of signal line and MACD line is considered for buy or sell decision.

As discussed in prior sections, when the MACD line crosses from below to above the signal line, it is considered as bullish, however when the MACD line crosses from above to below the signal line, the indicator is considered bearish. Therefore, following logic has been coded in the strategy:

- If MACD turns up from below the signal line to above signal line it implies bullish momentum, long order is generated
- If MACD turns below from above the signal line to become negative implies bearish momentum, short order is generated

Time-span for respective Shorter Period EMA, Longer Period EMA and Signal line has been calibrated based on the back-testing observations.

Using above mentioned logic trading strategy has been implemented in PyAlgoTrade framework (source code detail Appendix II).

8.6 Short Position Signal

As this research deals with Index Futures trading, its logical to utilize ease of short position available in Index Futures trading. In order to implement and track the strategy better, separate long and short position keeping as well as separate logic for long entry/exit and short entry/exit has been considered.

Like any algorithmic trading approach the goal has been to fully automate system with minimum need of monitoring and manual intervention. The switching of strategy based on regime switch has been automated in the implemented code and the position size is automated based on available cash at the execution point in time.

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The concluded plan for each of the Mean-Reversion regime and Momentum regime included logic for short entry and exit in addition to usual long entry and exit. This section describes the respective logic used in the concluded plan.

9. Trading Strategy Testing

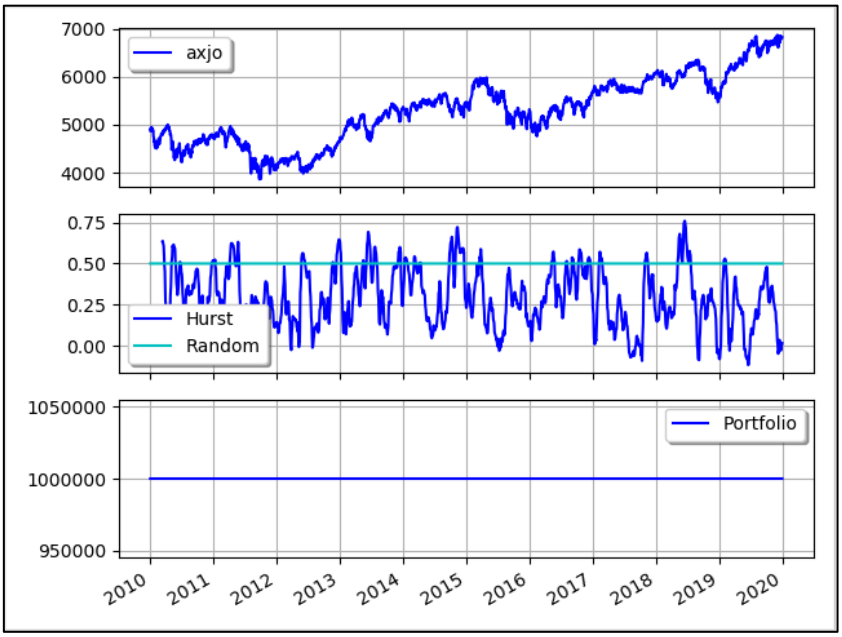
As discussed in previous sections, Equity Index future daily price data has been used for back-testing and framework PyAlgoTrade has been used as platform for back-testing. All the relevant future price data was downloaded in CSV format and fed into the python code developed using the framework PyAlgoTrade. This section of report presents the observation obtained from Hurst Exponent Analysis and the performance of trading strategy under development.

9.1. Moving Hurst Exponent Analysis Results

As discussed in previous sections, Look-back period for Moving Hurst Exponent analysis is significant. Empirical studies have concluded different behaviours for different look-back period. In this paper moving Hurst Exponent values was generated for range of look-back period starting from 20 to 120 days against daily price data for indices under investigation (Australia's ASX, South Korea's KOSPI, Hong Kong's HSCEI & HSI, Japan's Nikkei, India's Nifty, and Indonesia's JKSE). This section presents the snapshot of curves generated using python source code developed using PyAlgoTrade framework (source code detail Appendix II).

In Figure 1, first plot is the price performance of ASX. The second plot presents Moving Hurst Exponent of ASX with 50-days lookback period. Line representing Hurst value 0.5 (called Random in the plot) acts as divider for two the regime (Momentum and Mean-Reverting). Third plot, Portfolio is flat, as no strategy has been implemented yet.

Figure 1: Moving Hurst Exponent Analysis for Australia's ASX with 50-days look-back



created using PyAlgoTrade, top graph indicates price, middle graph moving Hurst exponent & bottom graph is a dummy portfolio

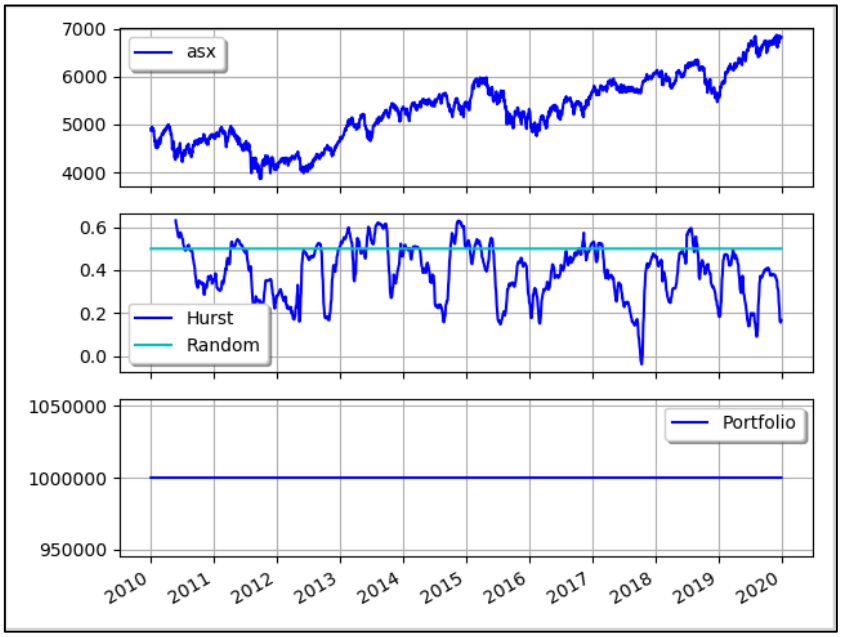
It can be concluded from above plot that moving Hurst Exponent value remains in mean-reversion regime (value < 0.5) for longer tenure compare to tenure in momentum regime (value > 0.5). This is

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an important observation as the strategy in this paper is based on market regime out of mean-reversion regime and momentum regime.

Similarly, for Figure 2, first plot is the price performance. The second plot presents Moving Hurst Exponent of Australia's ASX prices, but this time with 100-days lookback. Line representing Hurst value 0.5 (called Random in the plot) acts as divider for two the regime (Momentum and Mean-Reverting). Third plot, Portfolio is flat, as no strategy has been implemented yet.

Figure 2: Moving Hurst Exponent Analysis for Australia's ASX with 100-days look-back



created using PyAlgoTrade, top graph indicates price, middle graph moving Hurst exponent & bottom graph is a dummy portfolio

In line with what was observed for look back period 50-days, it can be viewed in plot obtained for 100-days look back period that moving Hurst Exponent value remains in mean-reversion regime (value < 0.5) for longer tenure compare to tenure in momentum regime (value > 0.5). Another important observation between the plot for 50 days look back period and plot for 100 days lookback period is that plot for 100-days lookback period is comparatively smooth whereas more zagged for 50-days lookback. It might be an important point to test for a trading strategy from stability point of view.

The analysis for other indices under investigation also reflects that the indices have Hurst Exponent properties and Moving Hurst Exponent indicators could be a valuable tool for trading these indices. It was observed that trend in values obtained for Hurst Exponent is consistent in the look-back periods from 50-days to 100-days. While overall trend in values remains consistent, as lookback period moves from 50-days to higher, curve become smoother. In other words, as it moves from 100 days to lower curve becomes move zagged. It also suggests that prices for indices are not random, it's either mean reverting or exhibiting trend or momentum behaviours.

Snapshot of other indices considered for this study is reflected on the next page (Figure 3 to Figure 14). The left panel reflects 50-day look back period. Whereas the right panel shows the performance of Hurst Exponent using 100-day lookback period. Like earlier, all graph outputs are created using PyAlgoTrade, top graph panel of each chart indicates price, middle panel in the graph shows the respective moving Hurst exponent & bottom graph in the panel is a dummy portfolio as no strategy has been initiated yet.

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Figure 3: Moving Hurst Exponent Analysis for South Korea's KOSPI with 50 days look-back

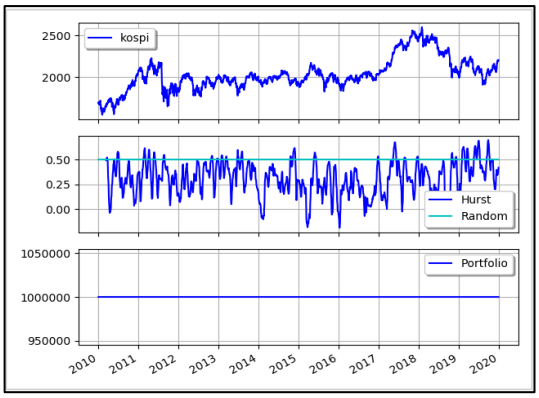


Figure 4: Moving Hurst Exponent Analysis for South Korea's KOSPI with 100 days look-back

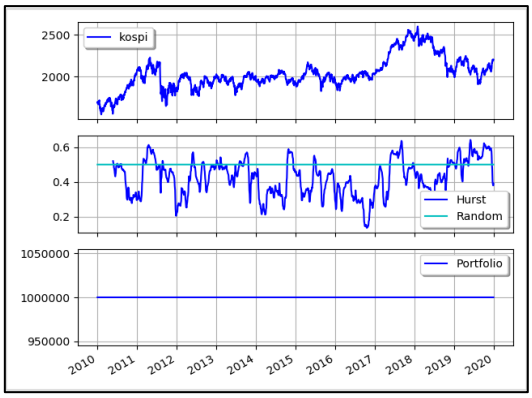


Figure 5: Moving Hurst Exponent Analysis for Hong Kong's HSCEI with 50 days look-back

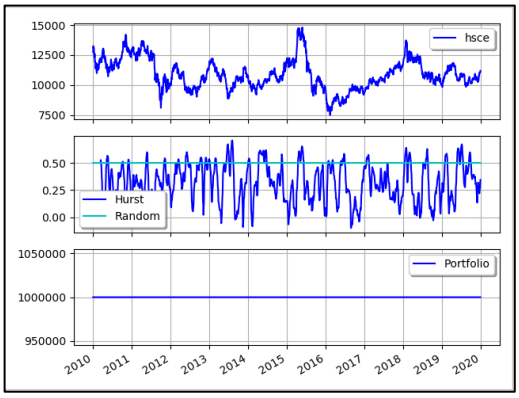


Figure 6: Moving Hurst Exponent Analysis for Hong Kong's HSCEI with 100 days look-back

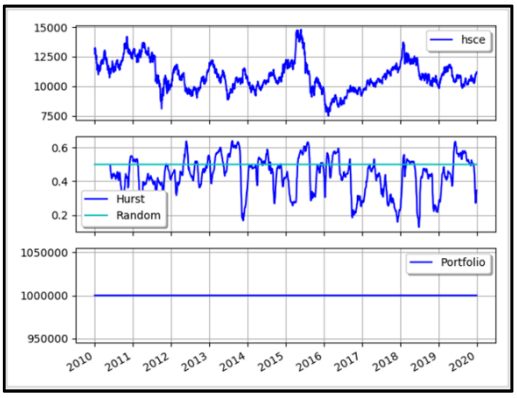


Figure 7: Moving Hurst Exponent Analysis for Hong Kong's HSI prices with 50 days look-back

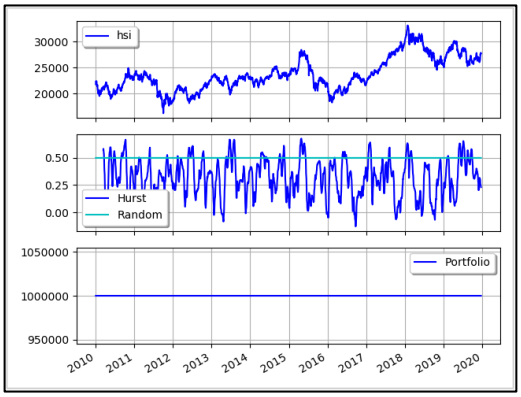


Figure 8: Moving Hurst Exponent Analysis for HSI prices with 100 days look-back

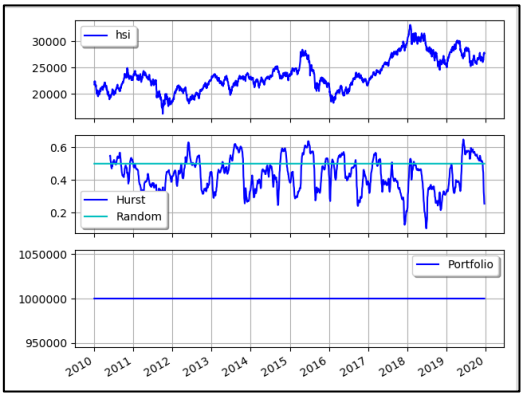


Figure 9: Moving Hurst Exponent Analysis for Japan's Nikkei prices with 50 days look-back

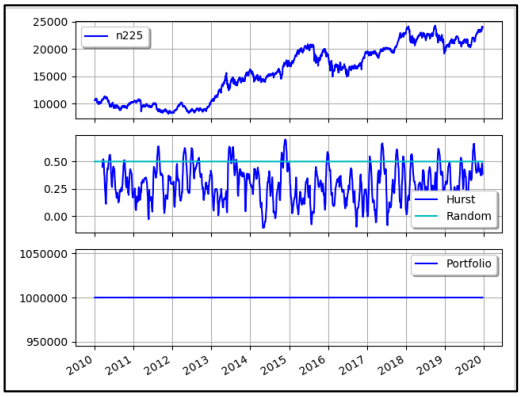
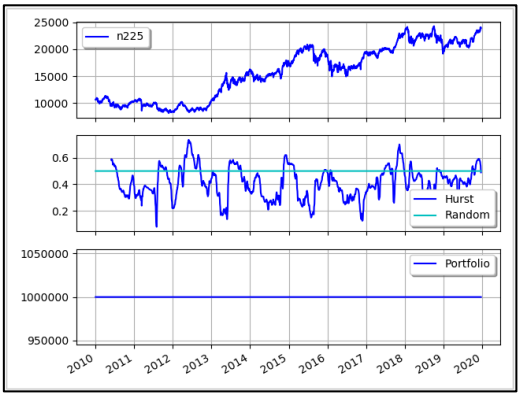


Figure 10: Moving Hurst Exponent Analysis for Japan's Nikkei prices with 100 days look-back



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Figure 11: Moving Hurst Exponent Analysis for India's Nifty prices with 50 days look-back

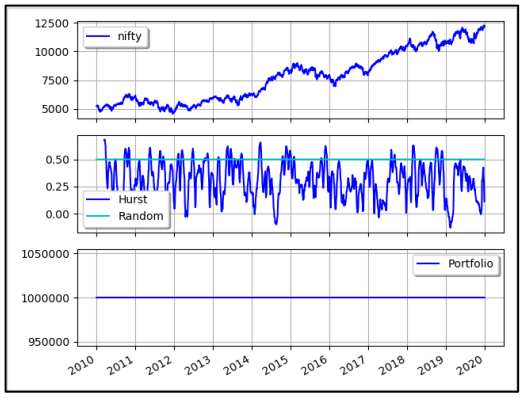


Figure 12: Moving Hurst Exponent Analysis for India's Nifty prices with 100 days look-back

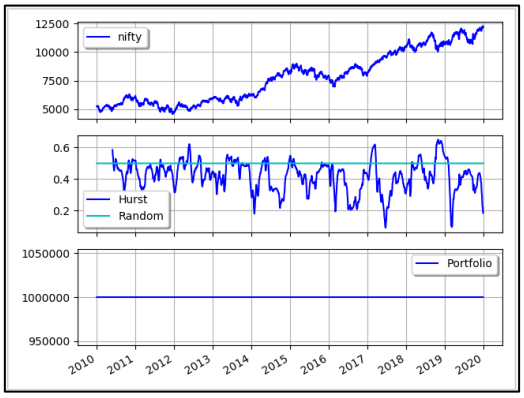


Figure 13: Moving Hurst Exponent Analysis for Indonesia's JKSE prices with 50 days look-back

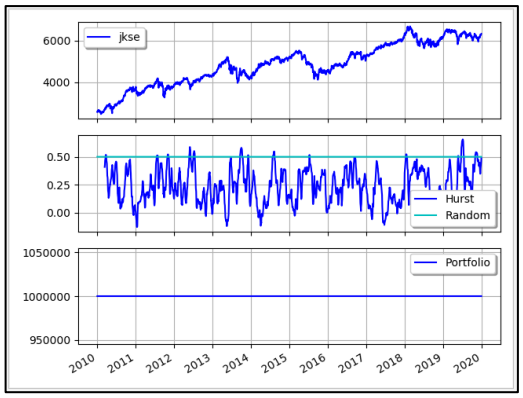
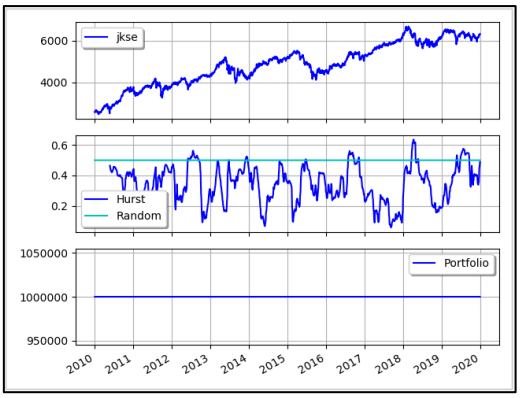


Figure 14: Moving Hurst Exponent Analysis for Indonesia's JKSE prices 100 days look-back



Since the Hurst exponent provides a measure for predictability, it can be used to guide data selection before forecasting. It also justifies the separate trading model built for periods with large Hurst Exponents (Hurst Exponents > 0.5) and period with lower Hurst exponent (Hurst Exponents < 0.5). These results can be used to forecast or predict the price behaviour at a pre-determined time interval in the future. Across all the indices analyse in this section, there is a common observation.

If Hurst exponent is used to determine the market regime out of Momentum regime and Mean-reversion regime, it's the Mean-reversion regime which is dominant. Therefore, a strategy covering mean-reversion regime of trading horizon strongly might perform well as per this theory.

In the next section of this report, Japan's Nikkei has been considered for analysis of performance for each of the trading strategy viz. Momentum and Mean-reversion regime. All other indices covered in this paper has been back-tested and analysed for consistency of evolved trading strategy but not presented here.

9.2 Trading Strategy Performance

This paper started with a logical concept to determine regime of market out of mean-reversion and momentum regime by making use of Hurst Exponent value and devise two sperate strategy the specific regime. This section presents the performance matrix observed for the respective strategies created during this paper. The start of 2010 to end of 2012 has been considered as Out-of-Sample period. The in-sample period for back-testing is from start of 2013 to end of 2019. All the matrix and plot has been captured using PyAlgoTrade framework.

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9.2.1 Performance of Attempted Approach

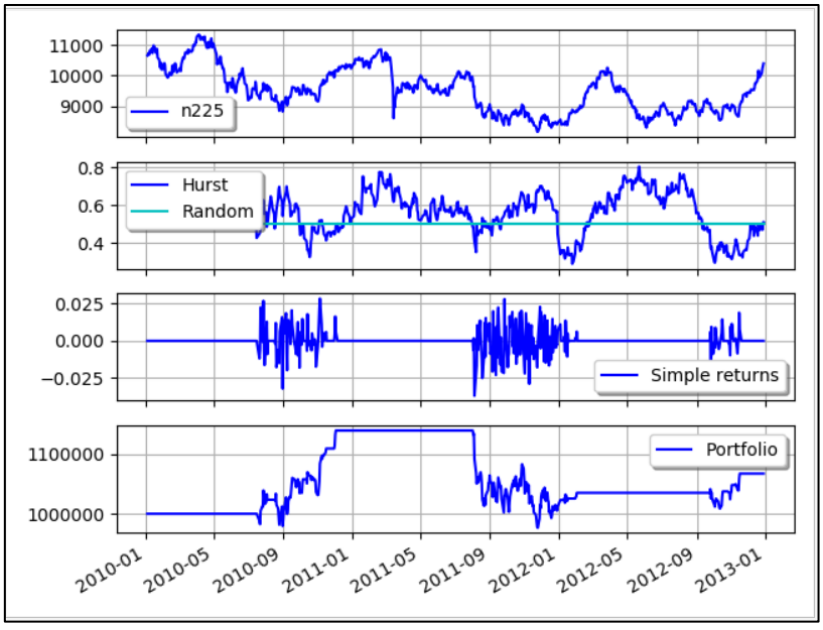
The evaluation of 'attempted approach' trading strategy follows the cross-validation approach and the entire data set (2010 to 2019) has been divided into two set 'in-sample period' and 'out-of-sample period'. Data set for each of the period has been back-tested independently. Start of 2010 to End of 2012 has been considered as 'out-of-sample' period and Start of 2013 to End of 2019 has been considered as 'in-sample' period. As Sharpe ratio, maximum drawdown and maximum drawdown period are ubiquitous as a comparative measures, these three-performance measures have been analysed for the trading strategy. Other common performance measures have been ignored due to limitations in PyAlgoTrade framework.

Table1: Parameters used in the 'attempted approach' trading strategy

Hurst Period	100
Momentum Regime Parameters	
Bollinger Bands Period	30
Bollinger Bands No of Standard Deviation	2
Slow SMA Period	40
Fast SMA Period	6
Mean Reversion Regime Parameters	
Standard Deviation Multiplier	0.2

Following is the performance of 'Attempted Approach' trading strategy for Japan's Nikkei in the respective period:

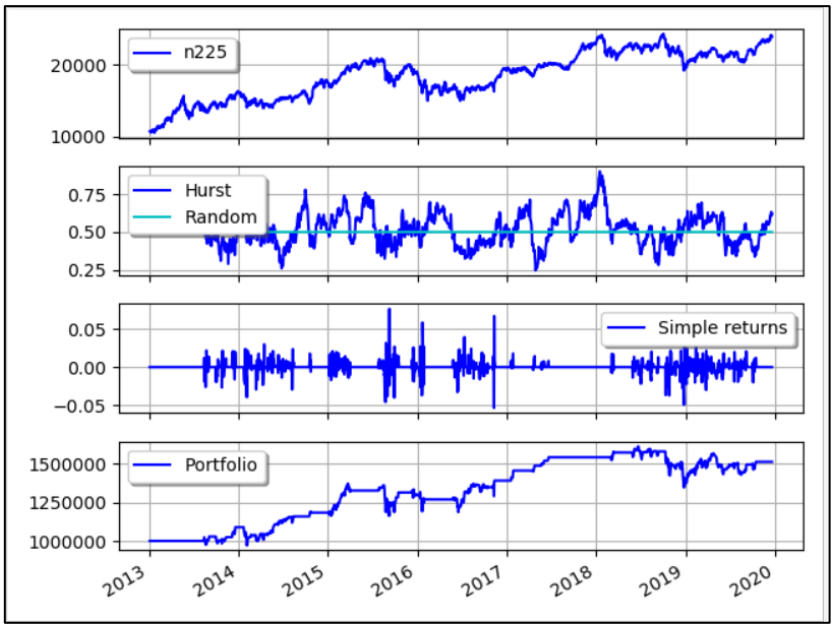
Figure 15: Plot obtained from strategy code for 'out-of-sample' period (2010 to 2012)



Sharpe ratio: -0.10
Maximum Drawdown: 0.19
Maximum Drawdown Period: 111 days
Cumulative returns: 31.19 %

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Figure 16: Plot obtained from strategy code for 'in-sample period' (2013 to 2019):



Sharpe ratio: 0.15
Maximum Drawdown: 0.17
Maximum Drawdown Period: 125 days
Cumulative returns: 51.39 %

As discussed in previous section, for a strategy to be robust, it's essential to generate consistent return in all different samples of test data.

From the performance matrix captured above, it's evident that performance of this strategy has not been consistent between 'in-sample' period and 'out-of-sample' period.

9.2.2 Performance of Concluded Plan

As discussed in last section, the paper follows the cross-validation approach and the entire data set (2010 to 2019) has been divided into 'in-sample' period' and 'out-of-sample' period. Start of 2010 to End of 2012 has been considered as 'out-of-sample' period and Start of 2013 to End of 2019 has been considered as 'in-sample' period. In line with performance evaluation of 'attempted approach' trading strategy, Sharp ratio, maximum drawdown and maximum drawdown period has been analysed for concluded trading strategy.

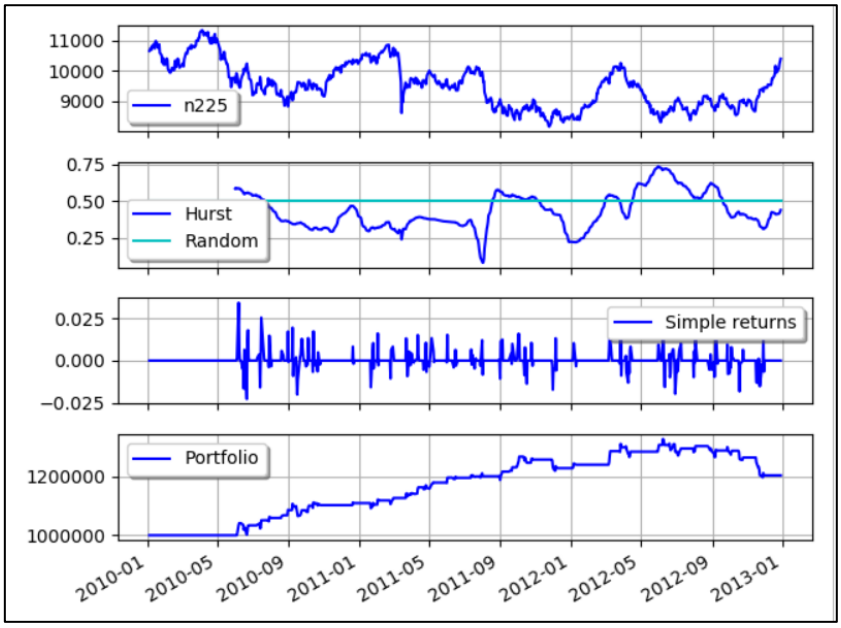
Table 2: Parameters used in the 'Concluded Plan' trading strategy

Hurst Period	100
Momentum Regime Parameters	
MACD Shorter Period	12
MACD Longer Period	26
MACD Signal Period	9
Mean Reversion Regime Parameters	
Entry SMA Period	80
Exit SMA Period	4
RSI Period	2
Overbought Threshold	80
Oversold Threshold	20

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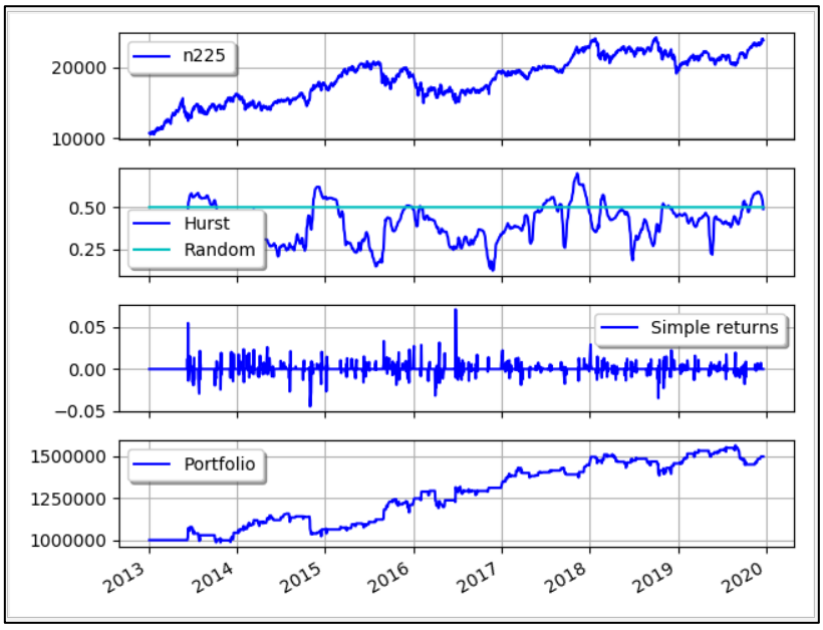
Following is the performance of 'Concluded Plan' trading strategy for Nikkei in the respective period:

Figure 17: Plot obtained from strategy code for 'out-of-sample' period (2010 to 2012)



Sharpe ratio: 0.22
Maximum Drawdown: 0.10
Maximum Drawdown Period: 45 days
Cumulative returns: 29.48%

Figure 18: Plot obtained from strategy code for 'in-sample period' (2013 to 2019)



Sharpe ratio: 0.27
Maximum Drawdown: 0.18
Maximum Drawdown Period: 62 days
Cumulative returns: 65.59 %

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Out of all parameters, look-back period for slow SMA (entry SMA) found to have maximum impact on Sharpe ratio (taking data for entire period – 2010 to 2019).

Table 3: Effects of Entry SMA

Entry SMA Period	55	60	65	70	75	80
Sharpe Ratio	0.12	0.15	0.16	0.26	0.23	0.13
Cumulative Return	49.02%	49.71%	50.83%	59.98%	56.88%	47.44%

A comparison between SMA vs EMA was done by replacing slow SMA with slow EMA and fast SMA with fast EMA while period was kept contact. It was found that SMA is more effective compare to EMA.

Rather than relying on only two set 'in-sample period' and 'out-of-sample period', a kind of k-fold cross validation has been attempted by dividing data into equally sized sub-sample. If a trading strategy is robust, the performance of smaller period when taken in separation would be consistent with entire period of back-testing. During this research the most effort has been invested in finding a solid strategy by experimenting different possible plan. As part of robustness test, performance analysis of concluded plan has been done for each of the 3-year period: 2010-2012, 2013-2015 and 2016-2018.

Table 4: k-fold cross validation for 3-year bucket interval

	2010-2012	**2013-2015**	**2016-2018**
Sharpe Ratio	0.22	0.24	0.26
Maximum Drawdown	0.10	0.16	0.15
Cumulative Return	29.48%	38.76%	45.87%

It was further analysed by adding more data in the specific bucket from in-sample period, but result remained approximately same.

Table 5: k-fold cross validation with more data for bucket intervals

	2010-2013	**2014-2016**	**2017-2019**
Sharpe Ratio	0.24	0.25	0.27
Maximum Drawdown	0.12	0.15	0.16
Cumulative Return	39.48%	48.76%	55.87%

As discussed in previous section, back-testing based trading strategy development is susceptible to many overfitting drawbacks. Therefore, its critical to validate the 'Concluded Plan' trading strategy with other liquid Asian indices. The back-testing for 'Concluded Plan' trading strategy has been performed for other liquid Asian indices in addition to Nikkei and it has been consistent. The next section of this report presents justification of various choices made during this paper.

10. Justification

This section of the presents justification of various choices made for this paper. As a part of this paper liquid Asian equity index futures have been chosen to develop a short-term trading model. Primary focus of this research is application of Hurst Exponent has been chosen for study. Back-testing methodology has been chosen to make this exercise more practical and application oriented. Later during the paper new indicators Moving Hurst Exponent and Moving Half-life of mean reversion were experimented.

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10.1 Justification for Trading Instrument

Leverage, liquidity, availability of short trading and lower costs are major benefits of Equity Index Futures. Very high volume of long and short trades is executed at any given time, therefore it's easier to place orders quickly without any adverse price movements, especially for futures contracts near maturity. These are exchange listed instruments and it has standardised clearing and settlement procedures. Having the exchange as an intermediary ensures fulfilment of all liabilities and the exchange also acts as a guarantor between transacting parties. Profits on futures are taxed at lower rate compared to short-term profits from physical stock. Keeping these advantages in mind index future has been considered for this research.

10.2 Justification for Hurst Exponent

The two most commonly used trading strategies in financial markets are trend-following and mean-reversion. Trends usually follow a longer time frame, but mean-reversion around the trend could be for a shorter time frame. As reflected earlier in the Literature Review section of this report, Hurst exponent provides a robust technique to classify market price returns in to either of the two strategies or if the prices are following just a random Brownian motion. Taking these distinguishing features in to account, Hurst Exponent has been chosen as main Quantitative method for this research.

10.3 Justification for Methodology

Back-testing is a practical method widely used by practitioners to measuring the performance of a trading strategy. The idea behind this methodology is very simple and intuitive as historical performances can be used to gauge the range of approximate future performance of a strategy. Back-testing allows various statistical validations and provides a way to refine and improve a trading strategy before deploying it. Keeping these benefits and ease of use in mind, this methodology has been chosen for this research.

10.4 Justification for Back-testing Framework

Paper makes use of open source, event driven Algorithmic Trading framework PyAlgoTrade for trading strategy design, evaluation and back-testing. This framework facilitates easy customization of back-test with historical data sourced from CSV files. Which makes it easier to experiment with different time periods by modifying the size of CSV files. It provides common interface for live-trading, paper-trading and back-testing with historical data. By design, it automates the algorithm's reaction to trade events and provides current or historical point-in-time data to eliminate look-ahead bias. (PyAlgoTrade documentation 0.20)

Framework also provides seamless integration with famous technical analysis library TA-Lib. It has easy to use functions for analysing the trading strategy that provides Performance metrics like Sharpe ratio and drawdown analysis (PyAlgoTrade documentation 0.20).

Apart from these advantages, PyAlgoTrade is easy to install and use. Initially during the paper another framework, Zipline was tried but it turned out very difficult to work with Historical data downloaded from investing.com. Keep all above factors in the prospective and time constraints, PyAlgoTrade was chosen for this paper.

11. Limitations & Further Study

Study is based on the assumption that new market information will generate a new trading signal resulting into trade execution. But in a live trading there may be many other unknown involved in this process. Transaction Costs, Slippage and Market Impact has been ignored in this study. The framework

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PyAlgoTrade has many known limitations and its very precise in terms of performance measures produced by it. A more comprehensive back-testing platform like Zipline or Back -trader that provides many more performance measures should be tried for further study. The research produce requires further analysis study on the dangers of overfitting, transaction cost and transaction constrain.

Research found that Moving Hurst Exponent based strategy was able to produce better return when compared with strategy without it. However, it must be stated that these profits observed in this research is purely theoretical. Usually, at least for small trading account there is a significant transaction cost and unfortunately Moving Hurst Exponent based strategy would generate relatively large number of transactions. However, transaction fees depend on the Index Future provider and a large volume trading firm can get away with these for free.

Research concludes from number of experimental strategies involving Hurst exponent that non-linear time series study is a complex topic. It requires more study from many different angles and not just simple Hurst exponent calculation of price time series. Hurst exponent for price return and price volatility should be explored to make trading strategy more robust.

12. Conclusion

The goal of this research was to develop and test Moving Hurst Exponent based trading strategy for liquid Asian index futures by making use of Hurst exponent of the price series. The outcome of this study exercise provides needed depth of knowledge of existing trading indicators in addition to Hurst Exponent analysis. This paper has documented the working experience with trading strategy development, use of back-testing cum trading framework and trading strategy evaluation experience. As part of new trading strategy development exercise, many well-known existing trading strategies were studied and back-tested with historical data.

Multiple combination of trading strategy were tried during the paper and two final comprehensive strategies named ‘attempted approach’ and ‘concluded plan’ has been presented in this report. The entry signal, exit signal and trade size etc. has been specified clearly in the respective strategies. As described in this report the resulting final ‘concluded plan’ out of this research paper as well ‘attempted approach’ has been back-tested with 10 years of historical daily price data. As stated in many academic literatures, 10 years’ worth of data would constitute sufficient number of observations to determine its success.

Research found that Hurst exponent can be used to determine the market regime out of Momentum regime and Mean-reversion regime. It was observed during the back testing that developing a trading strategy is essentially a task to determine whether prices for a certain time horizon will be mean-reverting or trending. Research further found that all the indices under investigation (Australia’s ASX, South Korea’s KOSPI, Hong Kong’s HSCEI & HSI, Japan’s Nikkei, India’s Nifty and Indonesia’s JKSE) exhibits dominant mean-reversion regime. Hurst Exponent value remain in mean-reversion regime (value < 0.5) for longer tenure compare to tenure in momentum regime (value > 0.5). This behaviour is consistent for all look-back period for Hurst Exponent calculation in the range 50 days to 100 days. Moving Hurst Exponent based strategy was able to produce better return when compared with strategy without it. Separate position keeping for long and short position was experimented for ‘concluded plan’ and it was found that independent signal execution against respective long and short position gives better performance. Research concludes that more research would be require to make long and short signals more robust.

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Appendix I: Asian Equity Index Futures

Below table summarizes settlement price mechanism for actively traded Asian equity index futures contract.

Table 6: Daily Settlement Price Mechanism

Exchange, Country	Country of the underlying	Contract	Active Contracts	Daily Settlement Price based on
ASX, Australia	Australia	ASX SPI 200	Quarterly	Last Traded Price
SGX, Singapore	China	Xinhua A50	Monthly	Closing Auction
CFFEX, China	China	CSI 300	Monthly	VWAP of last 60 minutes
HKEX, Hong Kong	Hong Kong	HSI	Monthly	Last Traded Price
HKEX, Hong Kong	Hong Kong	HSCEI	Monthly	Last Traded Price
SGX, Singapore	India	SGX Nifty	Monthly	Closing Auction
NSE, India	India	Nifty 50	Monthly	VWAP of last 30 minutes
JPX, Japan	Japan	Nikkei 225	Quarterly	Closing Auction
JPX, Japan	Japan	TOPIX	Quarterly	Closing Auction
BMD, Malaysia	Malaysia	FBM KLCI	Monthly	Last Traded Price
SGX, Singapore	Singapore	SiMSCI	Monthly	Closing Auction
KRX, South Korea	South Korea	Kospi 200	Quarterly	Closing Auction
SGX, Singapore	Taiwan	TaMSCI	Monthly	Closing Auction
TAIFEX, Taiwan	Taiwan	TAIEX	Monthly	VWAP of last 1 minute
TFEX, Thailand	Thailand	SET50	Quarterly	VWAP of last 5 minutes

Source: Respective Futures Exchange Websites

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Appendix II: Python Source Code

Source code of **Final Trading model**

MovingHurst.py

```
from pyalgotrade import technical
from hurst import compute_Hc
import numpy as np

class HurstExponentEventWindow(technical.EventWindow):
    def __init__(self, period, minLags, maxLags, logValues=True):
        super(HurstExponentEventWindow, self).__init__(period)
        self.__minLags = minLags
        self.__maxLags = maxLags
        self.__logValues = logValues

    def onNewValue(self, dateTime, value):
        if value is not None and self.__logValues:
            value = np.log10(value)
        super(HurstExponentEventWindow, self).onNewValue(dateTime, value)

    def getValue(self):
        ret = None
        if self.windowFull():
            ret = compute_Hc(self.getValues(), kind='price')[0]
        return ret

class HurstExponent(technical.EventBasedFilter):
    """Hurst exponent filter based on technical.EventBasedFilter
    :param dataSeries: The DataSeries instance
    :type dataSeries: :class:`pyalgotrade.dataseries.DataSeries`
    :param period: The number of values to use to calculate the hurst
exponent
    :type period: int
    :param minLags: The minimum number of lags to use. Must be >= 2
    :type minLags: int
    :param maxLags: The maximum number of lags to use. Must be > minLags
    :type maxLags: int
    :param maxLen: The maximum number of values to hold
        Once a bounded length is full, when new items are added, a
corresponding number of items are discarded
        from the opposite end. If None then dataseries.DEFAULT_MAX_LEN is
used
    :type maxLen: int
    """

    def __init__(self, dataSeries, period, minLags=2, maxLags=20,
logValues=True, maxLen=None):
        assert period > 0, "period must be > 0"
        assert minLags >= 2, "minLags must be >= 2"
        assert maxLags > minLags, "maxLags must be > minLags"

        super(HurstExponent, self).__init__(
            dataSeries,
            HurstExponentEventWindow(period, minLags, maxLags, logValues),
            maxLen
        )
```


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StrategyUtil.py

```
from pyalgotrade import strategy

from pyalgotrade.technical import rsi
from pyalgotrade.technical import ma
from pyalgotrade.technical import cross

from pyalgotrade.technical import macd

import MovingHurst

from pandas.plotting import register_matplotlib_converters
register_matplotlib_converters()


class ComprehensiveStrategy(strategy.BacktestingStrategy):
    def __init__(self, feed, instrument, hurstPeriod, macdShorterPeriod,
macdLongerPeriod, macdSignalPeriod,  rsiPeriod, entrySMAPeriod,
exitSMAPeriod,
                 overBoughtThreshold, overSoldThreshold):
        strategy.BacktestingStrategy.__init__(self, feed)

        self.__instrument = instrument
        self.__hurstPeriod = hurstPeriod

        # Use adjusted close values, if available, instead of regular close
values.
        if feed.barsHaveAdjClose():
            self.setUseAdjustedValues(True)
        self.__priceDS = feed[instrument].getPriceDataSeries()
        self.__adjClose = feed[instrument].getAdjCloseDataSeries()

        self.__macdShorterPeriod = macdShorterPeriod
        self.__macdLongerPeriod = macdLongerPeriod
        self.__macdSignalPeriod = macdSignalPeriod
        self.__macd = macd.MACD(self.__adjClose, self.__macdShorterPeriod,
self.__macdLongerPeriod, self.__macdSignalPeriod)

        self.__hurst = MovingHurst.HurstExponent(self.__adjClose,
hurstPeriod)

        self.__entrySMA = ma.EMA(self.__priceDS, entrySMAPeriod)
        self.__exitSMA = ma.EMA(self.__priceDS, exitSMAPeriod)

        self.__rsi = rsi.RSI(self.__adjClose, rsiPeriod)
        self.__overBoughtThreshold = overBoughtThreshold
        self.__overSoldThreshold = overSoldThreshold

        self.__longPos = None
        self.__shortPos = None


    def getHurst(self):
        return self.__hurst

    def getHurstValue(self):
        value = self.__hurst.getEventWindow().getValue()
        return value if value else 0.5
```


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```
    def onEnterOk(self, position):
        if self.__longPos == position:
            execInfo = position.getEntryOrder().getExecutionInfo()
            self.info("Enter Long order at $%.2f " % (execInfo.getPrice()))
        elif self.__shortPos == position:
            execInfo = position.getEntryOrder().getExecutionInfo()
            self.info("Enter Short order at $%.2f " %
(execInfo.getPrice()))

    def onEnterCanceled(self, position):
        if self.__longPos == position:
            self.__longPos = None
        elif self.__shortPos == position:
            self.__shortPos = None


    def onExitOk(self, position):
        if self.__longPos == position:
            execInfo = position.getExitOrder().getExecutionInfo()
            self.info("Exit Long order at $%.2f" % (execInfo.getPrice()))
            self.__longPos = None
        elif self.__shortPos == position:
            execInfo = position.getExitOrder().getExecutionInfo()
            self.info("Exit Short order at $%.2f" % (execInfo.getPrice()))
            self.__shortPos = None

    def onExitCanceled(self, position):
        # If the exit was canceled, re-submit it.
        position.exitMarket()

    def isMomentumRegimeEnterLongSignal(self):
        return cross.cross_above(self.__macd, self.__macd.getSignal()) > 0

    def isMomentumRegimeExitLongSignal(self):
        return cross.cross_above(self.__macd, self.__macd.getSignal()) < 0
and not self.__longPos.exitActive()

    def isMomentumRegimeEnterShortSignal(self):
        return cross.cross_below(self.__macd, self.__macd.getSignal()) > 0

    def isMomentumRegimeExitShortSignal(self):
        return cross.cross_below(self.__macd, self.__macd.getSignal()) < 0
and not self.__shortPos.exitActive()

    def onBars(self, bars):
        # Wait for enough bars to be available to calculate SMA and RSI.
        if self.__exitSMA[-1] is None or self.__entrySMA[-1] is None or
self.__rsi[-1] is None:
            return

        bar = bars[self.__instrument]
        hurst = self.getHurstValue()
        if hurst is None:
            self.meanRevRegimAlgo(bar)
        elif hurst < 0.5:
            self.meanRevRegimAlgo(bar)
        elif hurst > 0.5:
            self.meanRevRegimAlgo(bar)

    def momenRegimAlgo(self, bar):
        if self.__longPos is not None:
```


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```
            if self.isMomentumRegimeExitLongSignal():
                self.__longPos.exitMarket()
        elif self.__shortPos is not None:
            if self.isMomentumRegimeExitShortSignal():
                self.__shortPos.exitMarket()

        if self.isMomentumRegimeEnterLongSignal():
            shares = int(self.getBroker().getCash() * 0.9 / bar.getPrice())
            self.__longPos = self.enterLong(self.__instrument, shares,
True)

        if self.isMomentumRegimeEnterShortSignal():
            shares = int(self.getBroker().getCash() * 0.9 / bar.getPrice())
            self.__shortPos = self.enterShort(self.__instrument, shares,
True)

    def meanRevRegimAlgo(self, bar):
        if self.__longPos is not None:
            if self.isMeanRevExitLongSignal():
                self.__longPos.exitMarket()
        elif self.__shortPos is not None:
            if self.isMeanRevExitShortSignal():
                self.__shortPos.exitMarket()
        else:
            if self.isMeanRevEnterLongSignal(bar):
                shares = int(self.getBroker().getCash() * 0.9 /
bar.getPrice())
                self.__longPos = self.enterLong(self.__instrument, shares,
True)
            elif self.isMeanRevEnterShortSignal(bar):
                shares = int(self.getBroker().getCash() * 0.9 /
bar.getPrice())
                self.__shortPos = self.enterShort(self.__instrument,
shares, True)


    def isMeanRevEnterLongSignal(self, bar):
        return bar.getPrice() > self.__entrySMA[-1] and self.__rsi[-1] <=
self.__overSoldThreshold

    def isMeanRevExitLongSignal(self):
        return cross.cross_above(self.__priceDS, self.__exitSMA) and not
self.__longPos.exitActive()

    def isMeanRevEnterShortSignal(self, bar):
        return bar.getPrice() < self.__entrySMA[-1] and self.__rsi[-1] >=
self.__overBoughtThreshold

    def isMeanRevExitShortSignal(self):
        return cross.cross_below(self.__priceDS, self.__exitSMA) and not
self.__shortPos.exitActive()
```


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Main.py

```
from pyalgotrade import plotter

from pyalgotrade.stratanalyzer import sharpe
from pyalgotrade.stratanalyzer import returns
from pyalgotrade.stratanalyzer import drawdown
from pyalgotrade.stratanalyzer import trades

from pyalgotrade.barfeed import yahoofeed
from pandas.plotting import register_matplotlib_converters
register_matplotlib_converters()

import StrategyUtil

def main(plot):
    # Load  bar feed from CSV file/Yahoo format
    feed = yahoofeed.Feed()

    instrument = "n225"
    feed.addBarsFromCSV(instrument, r".\Data\n225.csv")

    #instrument = "hsi"
    #feed.addBarsFromCSV(instrument, r".\Data\hsi.csv")

    # instrument = "hsce"
    # feed.addBarsFromCSV(instrument, r".\Data\hsce.csv")

    # instrument = "tsec"
    # feed.addBarsFromCSV(instrument, r".\Data\tsec.csv")

    # instrument = "asx"
    # feed.addBarsFromCSV(instrument, r".\Data\asx.csv")

    # instrument = "kospi"
    # feed.addBarsFromCSV(instrument, r".\Data\kospi.csv")

    #instrument = "nifty"
    #feed.addBarsFromCSV(instrument, r".\Data\nifty.csv")

    #instrument = "jkse"
    #feed.addBarsFromCSV(instrument, r".\Data\jkse.csv")

    # parameters
    hurstPeriod = 100

    macdShorterPeriod = 12
    macdLongerPeriod = 36
    macdSignalPeriod = 9

    entrySMAPeriod = 65
    exitSMAPeriod = 4
    rsiPeriod = 2
    overBoughtThreshold = 80
    overSoldThreshold = 20

    strategy = StrategyUtil.ComprehensiveStrategy(feed, instrument,
hurstPeriod, macdShorterPeriod, macdLongerPeriod, macdSignalPeriod,
```


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```
rsiPeriod, entrySMAPeriod, exitSMAPeriod, overBoughtThreshold,
overSoldThreshold)

    #Attach a Sharpe Ratio analyser
    sharpeRatioAnalyzer = sharpe.SharpeRatio()
    strategy.attachAnalyzer(sharpeRatioAnalyzer)

    #Attach a Drawdown analyzer
    drawdownAnalyzer = drawdown.DrawDown()
    strategy.attachAnalyzer(drawdownAnalyzer)

    # Attach a return analyzer
    returnsAnalyzer = returns.Returns()
    strategy.attachAnalyzer(returnsAnalyzer)

    # Attach trade analyzer
    tradesAnalyzer = trades.Trades()
    strategy.attachAnalyzer(tradesAnalyzer)

    if plot:
        plt = plotter.StrategyPlotter(strategy, True, False, True)
        plt.getOrCreateSubplot("hurst").addDataSeries("Hurst",
strategy.getHurst())
        plt.getOrCreateSubplot("hurst").addLine("Random", 0.5)

        # Plot the simple returns on each bar.
        plt.getOrCreateSubplot("returns").addDataSeries("Simple returns",
returnsAnalyzer.getReturns())

    strategy.run()
    strategy.info("Final portfolio value: $%.2f" % strategy.getResult())
    print("Sharpe ratio: %.2f" % sharpeRatioAnalyzer.getSharpeRatio(0.05))
    print("Maximum Drawdown : %.2f" % drawdownAnalyzer.getMaxDrawDown())
    print("Longest Drawdown Duration : %s" %
drawdownAnalyzer.getLongestDrawDownDuration())
    print("Cumulative returns: %.2f %%" %
(returnsAnalyzer.getCumulativeReturns()[-1] * 100))

    print("")
    print("Total trades: %d" % (tradesAnalyzer.getCount()))
    if tradesAnalyzer.getCount() > 0:
        profits = tradesAnalyzer.getAll()
        print("Avg. profit: $%2.f" % (profits.mean()))
        print("Profits std. dev.: $%2.f" % (profits.std()))
        print("Max. profit: $%2.f" % (profits.max()))
        print("Min. profit: $%2.f" % (profits.min()))
        tradeAnalyzerAllReturns = tradesAnalyzer.getAllReturns()
        print("Avg. return: %2.f %%" % (tradeAnalyzerAllReturns.mean() *
100))
        print("Returns std. dev.: %2.f %%" % (tradeAnalyzerAllReturns.std()
* 100))
        print("Max. return: %2.f %%" % (tradeAnalyzerAllReturns.max() *
100))
        print("Min. return: %2.f %%" % (tradeAnalyzerAllReturns.min() *
100))

    print("")
    print("Profitable trades: %d" % (tradesAnalyzer.getProfitableCount()))
    if tradesAnalyzer.getProfitableCount() > 0:
        profits = tradesAnalyzer.getProfits()
        print("Avg. profit: $%2.f" % (profits.mean()))
```


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```
        print("Profits std. dev.: $%2.f" % (profits.std()))
        print("Max. profit: $%2.f" % (profits.max()))
        print("Min. profit: $%2.f" % (profits.min()))
        tradeAnalyzerPositiveReturns = tradesAnalyzer.getPositiveReturns()
        print("Avg. return: %2.f %%" % (tradeAnalyzerPositiveReturns.mean()
* 100))
        print("Returns std. dev.: %2.f %%" %
(tradeAnalyzerPositiveReturns.std() * 100))
        print("Max. return: %2.f %%" % (tradeAnalyzerPositiveReturns.max()
* 100))
        print("Min. return: %2.f %%" % (tradeAnalyzerPositiveReturns.min()
* 100))

    print("")
    print("Unprofitable trades: %d" %
(tradesAnalyzer.getUnprofitableCount()))
    if tradesAnalyzer.getUnprofitableCount() > 0:
        losses = tradesAnalyzer.getLosses()
        print("Avg. loss: $%2.f" % (losses.mean()))
        print("Losses std. dev.: $%2.f" % (losses.std()))
        print("Max. loss: $%2.f" % (losses.min()))
        print("Min. loss: $%2.f" % (losses.max()))
        tradeAnalyzerNegativeReturns = tradesAnalyzer.getNegativeReturns()
        print("Avg. return: %2.f %%" % (tradeAnalyzerNegativeReturns.mean()
* 100))
        print("Returns std. dev.: %2.f %%" %
(tradeAnalyzerNegativeReturns.std() * 100))
        print("Max. return: %2.f %%" % (tradeAnalyzerNegativeReturns.max()
* 100))
        print("Min. return: %2.f %%" % (tradeAnalyzerNegativeReturns.min()
* 100))

    if plot:
        plt.plot()

if __name__ == "__main__":
    main(True)
```


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Appendix III:

Code repository for above stated 'Final TradingModel':
https://github.com/vivekvision/TradingStrategy

Code repository for 'Other Attempted Strategies':
https://github.com/vivekvision/TradingStrategyTrial

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